

CHAPTER 3

Activities & Lab Packet

Force, Mass & Motion

A complete set of labs, worksheets, writing prompts, discussions, and group projects to accompany Chapter 3. Works with both the Standard and Easy Read versions of the textbook.



Georgia Standard S8P3

WHAT'S INSIDE

1	Inertia Egg Drop	LAB
2	Balloon Rocket Race	LAB
3	$F = ma$ in Action	LAB
4	Speed, Velocity, & Acceleration	WORKSHEET
5	Net Force Practice	WORKSHEET
6	Newton's Laws Sort	WORKSHEET
7	Writing Prompts (3 levels)	WRITING
8	Classroom Discussion Questions	DISCUSSION
9	Engineer a Bumper Car	GROUP
10	Forces & Motion Charades	REVIEW

1 Inertia Egg Drop

LAB

OBJECTIVE

Demonstrate Newton's First Law (inertia) with a dramatic visual.

TIME

25–35 minutes

STANDARDS

S8P3.b, S8P3.c

Materials (whole class demo + group practice)

- 1 plastic cup (per group)
- 1 plastic plate or stiff cardboard square (per group) — must be flat & smooth
- 1 cardboard tube (toilet paper roll)
- 1 plastic egg or ping-pong ball (NOT a real egg to start — too messy!)
- Water in the cup (fill 3/4 full)
- Paper towels (just in case)

Setup

Stack in this order from bottom up: cup of water → flat plate balanced on top of the cup → cardboard tube standing upright in the middle of the plate → plastic egg balanced on top of the tube. The egg should be directly above the cup.

Student Instructions

1. Predict: if you yank the plate out from the middle as fast as you can, what will happen to the egg?
2. Hit the side of the plate with a fast, horizontal swipe — straight out to the side. (Use a flat hand and HIT, don't pull slowly.)
3. The plate flies sideways. The tube flies sideways. But the egg falls STRAIGHT DOWN into the cup!
4. Why? Inertia! The egg was at rest. Nothing pushed the egg sideways (the plate was pulled out too fast for friction to drag the egg with it). So gravity pulls the egg straight down.
5. Try it 3 times. Did it work every time? Why might it fail?

TEACHER NOTES

Practice first: The trick is the SPEED of the hit. Slow = drag the cup over. Fast horizontal swipe = clean release. Practice once before students try.

Safety: Use plastic eggs or ping-pong balls. Real eggs are great for drama but make a huge mess. If a class has been very successful, the FINAL try can be a real egg as a reward.

The big idea: The egg has inertia — it wants to STAY at rest. Friction from the plate would drag it sideways if the plate moved slowly. A fast swipe doesn't give friction enough time to act.

Extension: The tablecloth trick is the same physics — yank a tablecloth fast and the dishes stay. Show a video clip of this if you have time.

Inertia Egg Drop — Observations

Name: _____

Period: _____

PREDICTION (before the demo):

What will happen to the egg when the plate is yanked out?

RESULTS:

Trial	Did it work? (egg in cup?)	Why or why not?
1		
2		
3		

Explain what happened using Newton's First Law (inertia):

Name ONE other example of inertia from your daily life:

2 Balloon Rocket Race

LAB

OBJECTIVE

Demonstrate Newton's Third Law (action-reaction).

TIME

30–40 minutes

STANDARDS

S8P3.b

Materials (per group of 2–3)

- 1 long balloon (regular balloons work too)
- 1 plastic drinking straw
- 3 meters of fishing line, kite string, or thin twine
- Masking tape
- 2 chairs or anchor points across the room
- Stopwatch (phone timer)
- Measuring tape

Setup

Thread the string THROUGH the straw before tying anything. Tie one end of the string to a chair. Pull the string tight across the room and tie it to another chair. The straw should slide freely along the string.

Student Instructions

1. Blow up the balloon. Pinch the end shut — do NOT tie it.
2. While holding the balloon closed, tape it to the straw so the opening faces backward.
3. Slide the straw to one end of the string.
4. Let go of the balloon. WATCH!
5. Measure how far the balloon traveled.
6. Time how long it took.
7. Try it 3 times. Then try a bigger balloon (or two balloons taped together).

The Big Question

The air rushes OUT the back of the balloon. So why does the balloon shoot FORWARD? Use Newton's Third Law to explain.

TEACHER NOTES

The key concept: The balloon pushes air BACKWARD (action). The air pushes the balloon FORWARD (reaction). This is exactly how a rocket works — rockets push hot gas down, the gas pushes the rocket up.

Troubleshooting: If the balloon spins instead of going forward, the tape isn't centered on the straw.

Extension competition: Award prizes for: longest distance, fastest time, most accurate (lands at the marked spot), most creative design.

Balloon Rocket Race — Data Sheet

Name: _____

Period: _____

Data:

Trial	Balloon size	Distance (m)	Time (sec)	Speed (m/s)
1				
2				
3				
4 (bigger)				

Identify the ACTION and REACTION:

Action: The balloon pushes _____

Reaction: The _____ pushes the balloon _____

Did the bigger balloon go faster, farther, or both? Why do you think so?

What's another real-world example of Newton's Third Law?

3 F = ma in Action

LAB

OBJECTIVE

Show that more force = more acceleration, and more mass = less acceleration.

TIME

30 minutes

STANDARDS

S8P3.b, S8P3.c

Materials (per group)

- 1 toy car or small wheeled cart (matchbox cars work great)
- 3 different sized rubber bands (small, medium, large)
- A few small heavy objects to add mass (coins, washers, or pennies in a small bag)
- Tape
- Measuring tape
- A long flat surface (floor or table)

Setup

Tape a rubber band to the back of the car. The rubber band acts as the "force." Stretch it back to a measured distance (say, 10 cm) and release. The further you stretch, the more force.

Student Instructions

- 1. Part A: Test FORCE.** Keep the car the same mass. Stretch the rubber band 5 cm and release. Measure how far the car goes. Now stretch it 10 cm. Then 15 cm. Record each distance.
- 2. Part B: Test MASS.** Keep the rubber band stretch the SAME (say, 10 cm). Add weight to the car (tape coins on top). Try the car with 0 coins, then 5 coins, then 10 coins. Measure distance each time.
- 3. Part C: Analyze.** When force went up (same mass), what happened to distance? When mass went up (same force), what happened to distance?

TEACHER NOTES

What students should find:

- More force (longer stretch) = more acceleration = more distance
- More mass (more coins) = less acceleration = less distance

This directly shows $F = m \times a$. With more force, acceleration goes up. With more mass and the same force, acceleration goes down.

Real-world tie-in: "Why is it easier to push an empty shopping cart than a full one? Same force, different mass, different acceleration."

Alternative: If no rubber bands, students can use a fan, a stretched balloon, or even a flick of the finger applied consistently.

F = ma in Action — Data Sheet

Name: _____

Period: _____

Part A: Change the FORCE (same car, no extra weight)

Rubber band stretch	Distance car traveled
5 cm	
10 cm	
15 cm	

When force went UP, the car went: ☐ farther ☐ shorter ☐ same

Part B: Change the MASS (same stretch, 10 cm)

Number of coins added	Distance car traveled
0 coins	
5 coins	
10 coins	

When mass went UP (same force), the car went: ☐ farther ☐ shorter ☐ same

Conclusion: Explain how this lab shows Newton's Second Law ($F = m \times a$):

4 Speed, Velocity, & Acceleration

WORKSHEET

OBJECTIVE

Distinguish between speed, velocity, and acceleration; do basic calculations.

TIME

20 minutes

STANDARDS

S8P3.a

Speed, Velocity & Acceleration Practice

Use the formulas: $\text{Speed} = \text{Distance} \div \text{Time}$ • $\text{Velocity} = \text{Speed} + \text{Direction}$

Name: _____

Period: _____

Part A: Is it speed, velocity, or acceleration? (Circle one)

1. "The car is moving 35 mph." → **Speed / Velocity / Acceleration**
2. "The runner sped up from 4 m/s to 8 m/s in 5 seconds." → **Speed / Velocity / Acceleration**
3. "The plane is flying 500 mph heading east." → **Speed / Velocity / Acceleration**
4. "The car slowed from 60 mph to 0 in 3 seconds." → **Speed / Velocity / Acceleration**
5. "The bike is going 12 mph north." → **Speed / Velocity / Acceleration**

Part B: Speed Calculations

6. A car drives 120 miles in 2 hours. What is its speed? _____
7. A runner covers 200 meters in 25 seconds. What is the runner's speed? _____
8. If a cyclist travels at 15 mph for 3 hours, how far did they go? _____
9. It takes 4 hours to drive 240 miles. What is the speed? _____

Part C: True or False (write T or F)

10. A car driving in a circle at a constant 30 mph is accelerating. _____
11. Speed and velocity are exactly the same thing. _____
12. Slowing down counts as acceleration. _____
13. On a distance-time graph, a steeper line means a faster speed. _____

ANSWER KEY

Part A: 1) Speed | 2) Acceleration | 3) Velocity | 4) Acceleration | 5) Velocity

Part B: 6) 60 mph | 7) 8 m/s | 8) 45 miles | 9) 60 mph

Part C: 10) T (changing direction = acceleration) | 11) F (velocity needs direction) | 12) T | 13) T

5 Net Force Practice

WORKSHEET

OBJECTIVE

Calculate net force; identify balanced vs. unbalanced forces.

TIME

15–20 minutes

STANDARDS

S8P3.b

Net Force Practice

Same direction = ADD. Opposite directions = SUBTRACT.

Name: _____

Period: _____

Calculate the NET force and state whether forces are BALANCED or UNBALANCED.

#	Situation	Net Force	Balanced or Unbalanced?
1	One person pushes a box right with 20 N. Another person pushes it right with 30 N.		
2	One person pushes a box right with 40 N. Another pushes left with 40 N.		
3	A 50 N force pulls a sled forward. Friction pulls back with 10 N.		
4	A book sits on a desk. Gravity pulls it down with 5 N. The desk pushes it up with 5 N.		
5	Two dogs play tug-of-war. One pulls right with 60 N. Other pulls left with 45 N.		
6	A car has 800 N pushing it forward. Air resistance is 800 N backward.		
7	Three kids push a wagon right: 15 N, 20 N, and 25 N. Friction pushes back 10 N.		
8	A soccer ball sits still in the grass. No one is touching it.		

Short answer:

What happens to an object's motion when net force is ZERO?

What happens to its motion when the net force is NOT zero?

ANSWER KEY

1) 50 N right — UNBALANCED | 2) 0 N — BALANCED | 3) 40 N forward — UNBALANCED | 4) 0 N — BALANCED | 5) 15 N right — UNBALANCED | 6) 0 N — BALANCED | 7) 50 N right — UNBALANCED | 8) 0 N — BALANCED

Short answer: If net force = 0, motion does NOT change (stays still OR keeps moving the same way). If net force $\neq$ 0, motion changes (speeds up, slows down, or changes direction).

6 Newton's Laws Sort

WORKSHEET

OBJECTIVE

Identify which of Newton's three laws applies to common scenarios.

TIME

15 minutes

STANDARDS

S8P3.b

Which Law Is It?

Write 1, 2, or 3 to show which Newton's Law applies to each example.

Name: _____

Period: _____

Law 1: Inertia

Things keep doing what they're doing unless a force acts on them.

Law 2: $F = ma$

Force = mass $\times$ acceleration. More force = more acceleration.

Law 3: Action/Reaction

Every action has an equal & opposite reaction.

Write 1, 2, or 3 in the blank:

#	Scenario	Law
1	A rocket pushes hot gas down and shoots upward.	
2	It's much harder to push a car than a bicycle with the same effort.	
3	When the bus stops suddenly, you fly forward.	
4	You pull a tablecloth fast — the dishes stay put.	
5	A swimmer pushes water backward, and the water pushes them forward.	
6	Kicking a soccer ball harder makes it fly faster.	
7	When you jump, you push down on Earth, and Earth pushes back up on you.	
8	A hockey puck slides on smooth ice and barely slows down.	
9	A bowling ball is much harder to throw at the same speed as a tennis ball.	
10	An astronaut floating in space drifts at the same speed forever (no friction).	
11	A bat hits a baseball — the ball flies away AND the bat pushes back on the batter's hands.	
12	If you push harder on a swing, the kid goes higher.	

Bonus: Make up your OWN example of each law.

Law 1: _____

Law 2: _____

Law 3: _____

ANSWER KEY

1) 3 | 2) 2 (more mass = harder to accelerate) | 3) 1 (your inertia keeps you moving) | 4) 1 | 5) 3 | 6) 2 (more force = more acceleration) | 7) 3 | 8) 1 (no force = stays moving) | 9) 2 (more mass needs more force) | 10) 1 | 11) 3 | 12) 2

7 Writing Prompts

WRITING

OBJECTIVE

Use scientific vocabulary to explain motion and force concepts in writing.

TIME

15–30 minutes each

STANDARDS

Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Day in the Life of a Force.** Pick one force you experienced today (pushing a door open, lifting a backpack, etc.). Describe what happened using the words "push," "pull," "force," and "motion."
2. **Why Seatbelts.** Explain why we wear seatbelts using what you know about inertia. Pretend you are explaining it to a younger kid.
3. **Fastest vs. Heaviest.** If a tiny mouse runs really fast and an elephant walks slowly, which one is harder to stop? Use the word "inertia" in your answer.

Level 2 — Intermediate (3 paragraphs)

4. **How a Rocket Works.** Use Newton's Third Law to explain how a rocket lifts off. Identify the action force and the reaction force. Where do they each act?
5. **Speed vs. Velocity vs. Acceleration.** Explain the difference between these three terms. Give a real-life example of each.
6. **The Bus Ride.** You're on a school bus. The bus speeds up, takes a curve, then brakes hard. Describe everything you feel during this ride using Newton's First Law.

Level 3 — Advanced (essay)

7. **The Driver's Test.** A new driver crashes their car into another car. Use Newton's Laws to explain: how their inertia caused them to keep moving forward, why the force they hit the other car with depended on their mass and speed, and how the action of their car hitting the other one made the other car move too. Be detailed!
8. **Sports and Newton.** Pick a sport you enjoy or watch. Explain how ALL THREE of Newton's Laws appear in that sport. Use specific examples (a play, a movement, a piece of equipment).

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE

Practice scientific thinking through verbal reasoning.

USE

Warm-ups, exit tickets, Socratic seminar

STANDARDS

Cross-cutting

TEACHER NOTES

Use think-pair-share format. Several questions are intentionally counterintuitive — they're great for sparking debate. Push students to defend with evidence, not just opinion.

Quick Warm-Ups (1–2 min each)

1. If a hockey puck slides forever on perfect ice, is any force needed to keep it moving?
2. If you sit perfectly still in a car going 60 mph, are YOU moving 60 mph?
3. Can you push something without it pushing back?
4. Why is it harder to stop a fully loaded truck than an empty one moving the same speed?
5. A baseball and a bowling ball — which has more inertia?

Deeper Discussions (5–15 min each)

6. **The Frictionless World.** Imagine you woke up tomorrow and friction didn't exist. What three things in your daily routine would be totally impossible? Why?
7. **Walking Is Impossible (Without Newton's Third Law).** Discuss: when you walk, what does your foot push against? What pushes you forward? Why can't you walk on perfectly slippery ice?
8. **The Equal Force Paradox.** When you push a wall with 30 N, the wall pushes you back with 30 N. So why don't they cancel out? Why doesn't anything happen?
9. **Astronaut on a Spacewalk.** An astronaut floats in space, far from any spaceship or planet. They throw a wrench. Describe what happens to BOTH the wrench AND the astronaut. Use Newton's Third Law.
10. **Big Trucks & Small Cars.** In a crash between a big truck and a small car, why is the small car usually more damaged — if the forces on each are equal (Newton's Third Law)?

9 Engineer a Bumper Car

GROUP PROJECT

OBJECTIVE

Design a "bumper" that protects an egg or fragile cargo during a collision.

TIME

2–3 class periods

STANDARDS

S8P3.b, S8P3.c

Project Description

Each group designs and builds a "vehicle" that protects an egg (or another fragile item) when it crashes into a wall at the bottom of a ramp. The challenge applies all three of Newton's Laws — especially inertia and action-reaction at impact.

Requirements

- The cargo (egg / plastic egg / ping-pong ball) must be visible and reachable
- The vehicle must roll freely down a ramp under gravity alone
- The vehicle must be no larger than 20 cm × 20 cm
- The egg must survive a crash from a 1-meter-high ramp into a wall

Easy Materials

Material	Use
Cardboard, paper towel tubes	The body of the vehicle
Bottle caps, plastic lids, old CDs	Wheels
Straws, BBQ skewers, pencils	Axles
Cotton balls, bubble wrap, foam, sponges	Crash protection (cushioning)
Tape, glue, rubber bands	Assembly

Required Deliverables

1. **Sketch the design** BEFORE building. Label which parts handle inertia, force, and the reaction at impact.
2. **Build and test.** Iterate at least once after the first crash test.
3. **Final crash test** in front of the class.
4. **Write-up:** Explain how Newton's First Law (egg's inertia tries to keep moving forward at impact), Second Law (more mass to slow down = more force needed), and Third Law (egg pushes on the bumper, bumper pushes on the egg) all show up in your design.

TEACHER NOTES

The big learning: At impact, the vehicle stops quickly — but the egg's inertia wants to keep moving. The bumper has to slow the egg down GRADUALLY (over a longer time) to reduce force. This is exactly how car airbags and crumple zones work in real life.

Tip: Have the ramp set up at a CONSISTENT angle — all groups test from the same height and slope so it's fair.

Rubric (suggested): Egg survives (30%), design creativity (20%), labels & physics explanation (30%), presentation (20%).

Real-world tie-in: This is exactly the engineering inside every modern car. Show a quick video of a real crash test if you have time.

10 Forces & Motion Charades

REVIEW GAME

OBJECTIVE

Review chapter vocabulary in an active, kinesthetic way.

TIME

20–30 minutes

STANDARDS

S8P3 cross-cutting

How to Play

Print and cut out the cards below. Divide the class into two teams. One person draws a card and acts out the term WITHOUT speaking. Their team has 60 seconds to guess. If they guess correctly, they earn a point. Most points at the end wins.

Variations

- **Pictionary mode:** Draw the term on the board instead of acting it out.
- **One Word Hint:** Allow the performer to give ONE word as a clue (but not the actual term).
- **"\$25,000 Pyramid" mode:** Performer gives clues about the term while their team guesses (no acting).

Charades Card Set — Print & Cut

MOTION	SPEED	VELOCITY
ACCELERATION	DECELERATION	FORCE
PUSH	PULL	FRICTION
INERTIA	MASS	WEIGHT
GRAVITY	BALANCED FORCES	UNBALANCED FORCES
NET FORCE	NEWTON'S 1ST LAW	NEWTON'S 2ND LAW
NEWTON'S 3RD LAW	ACTION FORCE	REACTION FORCE
AIR RESISTANCE	FREE FALL	NEWTON (N)

TEACHER NOTES

Easy adaptation: For struggling readers, allow the actor to whisper the word to a partner first, then act it out together.

Quick prizes: Homework pass, candy, classroom job of choice.

Time saver: Set a 60-second timer on your phone or projector and play through it continuously.

Easier-to-act terms first: Start with PUSH, PULL, GRAVITY, FRICTION (easy to mime), then work up to the abstract ones like INERTIA and NET FORCE.